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COMPUTER

CHAPTER 29

# Mobile Phones

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COMPUTER · CHAPTER 29

# Mobile Phones

## ਮੋਬਾਈਲ ਫੋਨ

IN THIS CHAPTER


- Important Inventors And Historical Facts
- How Mobile Phones Work
- Cellular Network Concept
- Multiple Access Techniques
- Mobile Phone Identification Codes
- Comparison Of Mobile Codes


A **Mobile Phone (Cell Phone)** is an electronic communication device used to make and receive calls using **radio frequency signals (electromagnetic waves)** while the user is moving within a cellular network coverage area.

Mobile phones communicate through a network of specialized base stations called **Cell Sites (Mobile Towers)**.

A mobile phone is an electronic device used for mobile telecommunications over a cellular network, allowing users to communicate through voice calls, messages, and internet services while moving from one location to another.

Key Features of Mobile Phones

Modern mobile phones support:

- Voice Calls
- SMS (Short Message Service)
- MMS (Multimedia Messaging Service)
- Email
- Internet Access
- Bluetooth
- Camera & Video Recording
- GPS Navigation
- Music Player
- Radio
- Mobile Banking
- Social Media Applications

Exam Point

Because mobile phones use a cellular network architecture, they are also called:

- Cellular Phones
- Cell Phones

## IMPORTANT INVENTORS AND HISTORICAL FACTS

| Invention | Inventor/Organization | Year |
|---|---|---|
| Walkie-Talkie | Donald Hings | 1937 |

| Invention | Inventor/Organization | Year |
|---|---|---|
| First Mobile Phone | Martin Cooper | 1973 |
| First Commercial Mobile Phone | Motorola DynaTAC 8000X | 1983 |
| First Smartphone | IBM Simon Personal Communicator | 1994 |
| First Cellular Network | Nippon Telegraph and Telephone (NTT), Japan | 1979 |

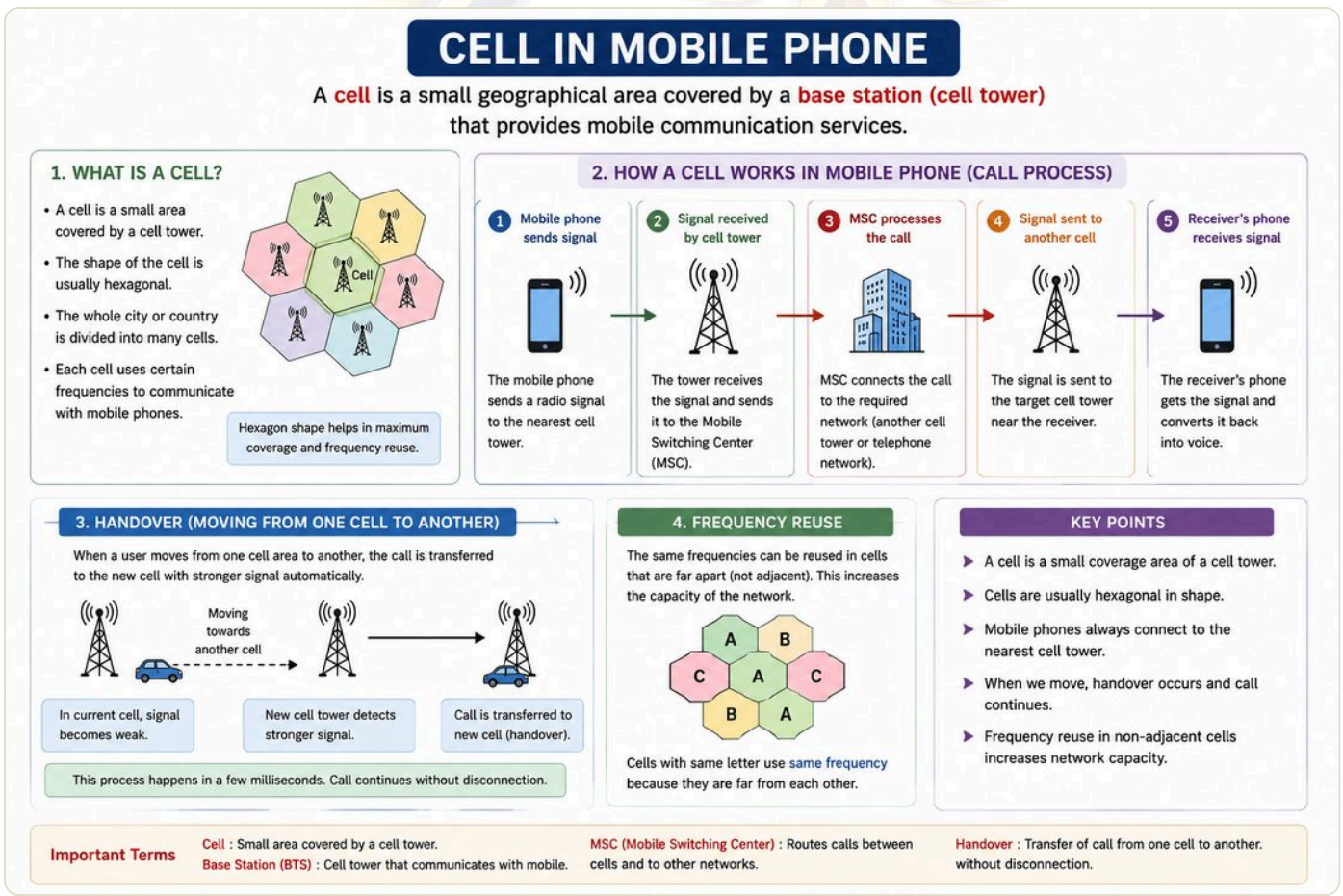
Exam Point :Martin Cooper is known as the "Father of the Mobile Phone".

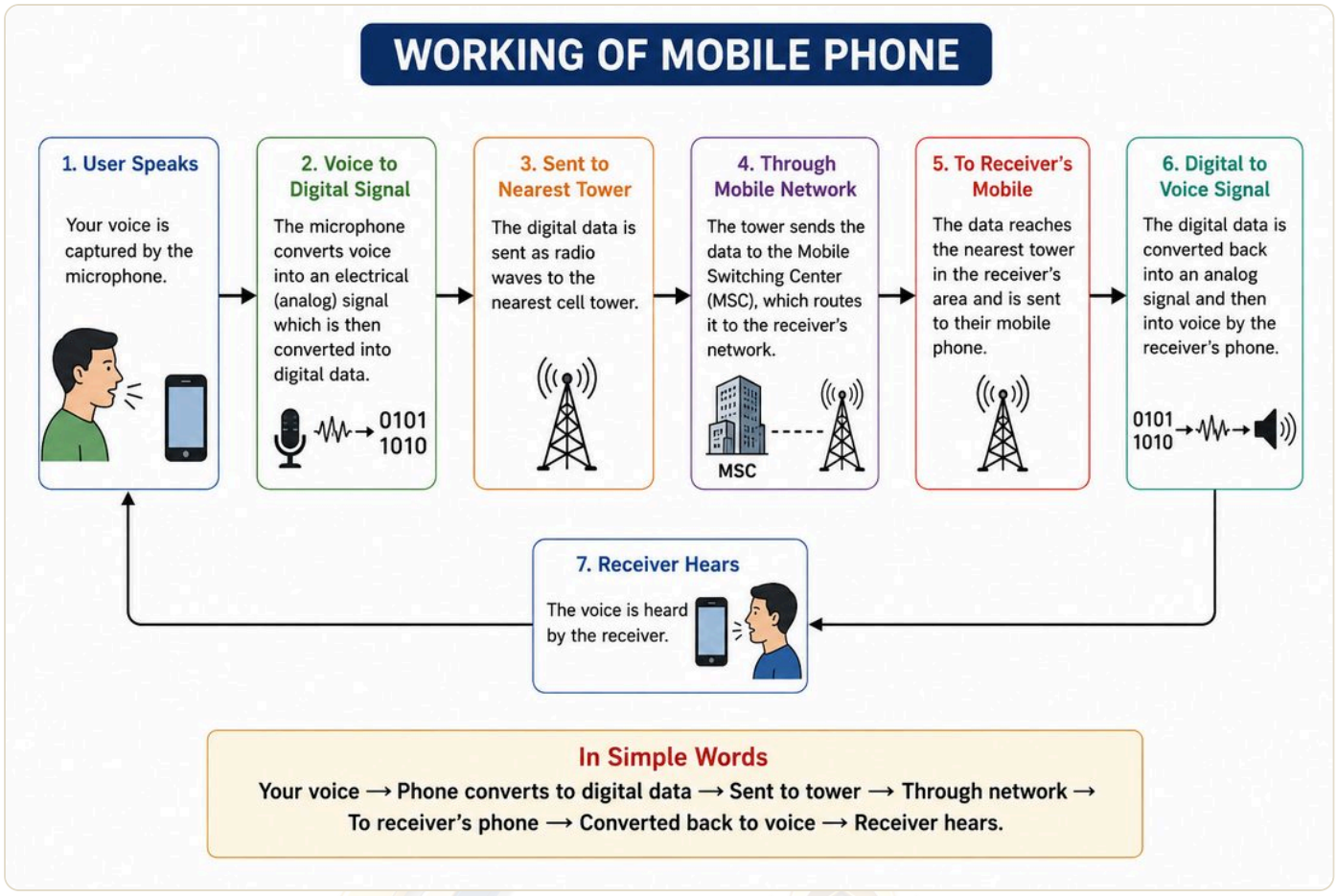
## HOW MOBILE PHONES WORK

Mobile phones communicate using **Radio Waves**.

A mobile phone converts voice into electrical signals and transmits them to nearby cell towers using radio frequencies.

## CELLULAR NETWORK CONCEPT






A large geographical area is divided into small regions called **Cells**.

Each cell is served by a base station (cell tower).

### Shape of Cell

For network planning purposes, the cell is assumed to be HEXAGON

### Full Duplex Communication

Mobile phones support Full Duplex Communication.

Both users can speak and listen simultaneously.

## MULTIPLE ACCESS TECHNIQUES

To allow many users to communicate simultaneously, cellular systems use multiple access techniques.

### 1. FDMA (Frequency Division Multiple Access)

Each user gets a separate frequency channel.

Features

- Different frequencies assigned to different users.
- Neighboring cells use different frequencies.

Example

User A → Frequency 1

User B → Frequency 2

User C → Frequency 3

2. TDMA (Time Division Multiple Access)

Multiple users share the same frequency but at different time slots.

Features

- Frequency divided by time.
- Each user transmits in a specific time period.

3. CDMA (Code Division Multiple Access)

All users share the same frequency but use unique codes.

Features

- More efficient than FDMA.
- Better capacity and security.

4. PDMA (Polarization Division Multiple Access)

Uses different polarization states of radio waves to separate signals.

# MOBILE PHONE IDENTIFICATION CODES

Every mobile phone contains special identification numbers.

1. ESN (Electronic Serial Number)

- Unique 32-bit number
- Permanently programmed into the phone

Exam Point

ESN is permanent.

2. MIN (Mobile Identification Number)

- 10-digit number
- Derived from the mobile number

3. SID (System Identification Code)

- Unique 5-digit number
- Assigned to each cellular carrier

Example

Jio, Airtel, Vi

## 4 The IMEI (International Mobile Equipment Identity)

IT is a unique 15-digit number assigned to every mobile device.

It serves as an identification code that differentiates each handset from others.

This number is crucial for several reasons:

- Device Identification: Uniquely identifies each handset on the network.
- Network Security: Mobile networks use IMEI to allow or block devices from accessing the network.
- Theft Prevention: If a phone is lost or stolen, the IMEI can be blacklisted, preventing its use with any SIM card.
- Warranty and Service: Used by manufacturers to verify warranty and service records of the device.
- You can usually find your IMEI by dialling *#06# on the keypad or checking the device's settings and packaging.

# COMPARISON OF MOBILE CODES

| Code | Full Form | Purpose |
|---|---|---|
| ESN | Electronic Serial Number | Identifies Phone |
| MIN | Mobile Identification Number | Identifies Subscriber |
| SID | System Identification Code | Identifies Network Provider |

# SIM CARD

SIM stands for: Subscriber Identity Module

A SIM card is a smart card issued by a mobile service provider(AIRTEL,JIO) that stores subscriber information.

Information Stored in SIM

- IMSI
- Phone Number
- Authentication Information
- Network Details

# IMSI

Full Form :International Mobile Subscriber Identity

A unique number stored in the SIM card used to identify the subscriber.

# TMSI

Full Form :Temporary Mobile Subscriber Identity

After IMSI verification, the network assigns a temporary identity called TMSI.

Purpose

- Improves privacy
- Protects IMSI from exposure

# ESIM

Full Form :Embedded Subscriber Identity Module

Features

- Built into device hardware
- No physical SIM card required
- Can be activated digitally

Exam Point eSIM = Embedded SIM.

# WHAT HAPPENS WHEN WE MAKE A CALL?

Step 1

Phone searches for a nearby control channel.

If unavailable:

**"No Service"** is displayed.

Step 2

Phone checks SID.

If SID matches:

Phone identifies its home network.

Step 3

Phone sends a registration request to the network.

Step 4

MTSO(Mobile telephone switching office ) records the phone's location.

Step 5

MTSO assigns a frequency pair.

Step 6

Call is connected.

Step 7

If user moves into another cell:

Network performs Handoff (Handover).

Call continues without interruption.

### MTSO (Mobile Telephone Switching Office)

Functions

- Call Routing
- Location Tracking
- Frequency Allocation
- Handover Management

Exam Point :MTSO is the central controller of cellular communication.

# HANDOFF (HANDOVER)

The process of transferring an ongoing call from one cell tower to another without disconnecting the call.

Example: While traveling in a car, your call continues because handoff occurs automatically

# IMPORTANT MOBILE COMMUNICATION TERMS

| Term | Full Form |
|---|---|
| Wi-Fi | Wireless Fidelity |
| WiMAX | Worldwide Interoperability for Microwave Access |
| WAP | Wireless Application Protocol |
| GPS | Global Positioning System |
| WCDMA | Wideband Code Division Multiple Access |
| GSM | Global System for Mobile Communications |
| LTE | Long Term Evolution |
| VoLTE | Voice over LTE |
| CDMA | Code Division Multiple Access |

# GSM SYSTEM

Full Form :Global System for Mobile Communications

Features

- Most widely used mobile communication standard
- Digital system
- Considered a 2G technology

## GSM CHANNELS

**Control Channel :Used for network management and signaling.**

Traffic Channel :Used for voice and data transmission.

# OPERATING SYSTEMS IN MOBILE PHONES

## ANDROID

Based On :Linux Kernel

Developed By :Google

## I O S

Developed By :Apple Inc.

Based On :Darwin Operating System

Exam Point

Android → Linux Kernel iOS → Darwin

# INTERNET SERVICE PROVIDERS (ISP)

### Telenet

World's First ISP

Established in 1974.

### VSNL (Videsh Sanchar Nigam Limited)

Introduced Internet services in India in 1995. It is India's First ISP

Exam Point :VSNL = First Internet Service Provider in India.

### PHUBBING

Ignoring a person physically present with you because you are busy using your smartphone.

# GENERATIONS OF MOBILE NETWORKS

Evolution of Mobile Communication

0G

2.75G

|

|

1G

3G

|

|

2G

4G

|

|

2.5G

5G

|

# 0G (ZERO GENERATION)

Features

- Pre-cellular mobile communication

- Radio Telephones
- Installed in vehicles

Services :Voice only

## 1G (FIRST GENERATION)

Introduced :1979

Country :Japan

Organization :NTT(Nippon telephone and telegraph)

Features

- Analog communication
- Voice only
- Low quality

Speed :2.4 kbps

Exam Point: 1G = Analog Technology

## 2G (SECOND GENERATION)

Features

- Digital technology
- Voice Calls
- SMS
- Email

Speed :64 kbps

Exam Point :2G introduced SMS.

## 2.5G

Technology :GPRS (General Packet Radio Service)

Features

- Packet-switched data
- Internet Access

## 2.75G

Technology :EDGE (Enhanced Data Rates for GSM Evolution)

Speed :Up to 128 kbps

Features

- Faster Internet

# 3G (THIRD GENERATION)

Features

- High-Speed Internet
- Video Calling
- Multimedia Services

Technologies

- UMTS
- WCDMA

Speed :Up to 2 Mbps

Exam Point :3G introduced practical mobile internet and video calling.

# 4G (FOURTH GENERATION)

Features

- IP-based Network
- High-speed broadband
- HD Streaming
- Online Gaming

Technologies

- LTE (Long Term Evolution)
- VoLTE(Voice over LTE)

Speed :Up to 100 Mbps

Exam Point :VoLTE supports voice and data simultaneously.

# 5G (FIFTH GENERATION)

Features

- Ultra-high speed
- Low latency
- Massive device connectivity
- Enhanced security

Applications

- Smart Cities
- IoT (Internet of Things)
- Self-driving Cars
- Remote Surgery
- Artificial Intelligence

Speed :Can exceed 1 Gbps under ideal conditions.

Exam Point

5G is approximately 10–30 times faster than 4G depending on network conditions.

## MOST IMPORTANT EXAM FACTS

1. Martin Cooper invented the mobile phone.
2. Motorola DynaTAC 8000X was the first commercial mobile phone.
3. IBM Simon was the first smartphone.
4. Hexagon is the ideal cell shape.
5. Mobile phones use radio waves.
6. Google Android is based on Linux Kernel.
7. iOS is based on Darwin.
8. GSM stands for Global System for Mobile Communications.
9. VSNL was India's first ISP.
10. Telenet was the world's first ISP.
11. IMSI is stored in the SIM card.
12. TMSI is a temporary identity.
13. MTSO controls call switching.
14. Handoff transfers calls between cells.
15. 1G = Analog.
16. 2G = Digital + SMS.
17. 3G = Mobile Internet.
18. 4G = LTE & VoLTE.
19. 5G = Ultra-fast communication.
20. Phubbing = Phone + Snubbing.

## QUICK REVISION NOTES

- Mobile phones use radio waves.
- Hexagon is the shape of a cellular cell.
- Martin Cooper invented the mobile phone.
- Motorola DynaTAC 8000X = First commercial mobile phone.
- IBM Simon = First smartphone.
- Android = Linux Kernel.
- iOS = Darwin.
- SIM = Subscriber Identity Module.
- IMSI = International Mobile Subscriber Identity.

- eSIM = Embedded SIM.
- GSM = Global System for Mobile Communications.
- Wi-Fi = Wireless Fidelity.
- GPS = Global Positioning System.
- MTSO controls mobile call routing.
- Handoff transfers calls between cells.
- 1G = Analog.
- 2G = SMS.
- 3G = Mobile Internet.
- 4G = LTE.
- 5G = Ultra-fast connectivity.